U5: The Forced Structure of Translation Symmetry and the Momentum Operator

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Abstract

We present the structural derivation of the spatial translation generator and the momentum operator on the participation-measure Hilbert space from the Event-Density (ED) primitive stack and the now-FORCED upstream items U1, U2-Discrete, U2-Continuum, and Theorem 10 (Born). Operating strictly outside Framing 1 — i.e., **without invoking U3** in any role, derivation input or otherwise — we decompose U5 into three sub-features: existence of a strongly continuous one-parameter unitary representation of spatial translations on the U2 Hilbert space (F1); uniqueness of its self-adjoint generator (F2); and identification of that generator with the standard momentum operator $\hat{p} = -i\hbar\nabla$, with plane waves as eigenfunctions and the standard L^2 Fourier transform as the unique unitary intertwiner of position and momentum representations (F3). F1 follows from Primitive 03 (graph homogeneity) and Primitive 06 (admissibility of translations along the spatial axis), with unitarity inherited from the U2 inner product. F2 follows from Stone's theorem on one-parameter unitary groups. F3 follows from the position-representation form of the translation generator together with the standard Fourier theory on $L^2(\mathbb{R}^d)$, with the principal alternatives — fractional-Fourier, wavelet, and Mellin transforms — explicitly dismissed by structural inconsistency with translation-driven conjugacy. All four falsifiers are dispatched. The U5 verdict is **forced**, with no new structural CANDIDATEs introduced and no dependence whatever on the sibling CANDIDATE U3. The arc closes the kinematic backbone of the QM-emergence Phase-1 program: U5 supplies the momentum operator that U4's Galilean integration acts upon, and the Fourier-conjugacy that Phase-1 Step 5 needs to derive the Heisenberg bound $\Delta x \cdot \Delta p \geq \hbar/2$. The U5 derivation is what makes U4's "form-FORCED–value-INHERITED" closure possible while preserving acyclicity. The active upstream-CANDIDATE inventory is unchanged by U5 alone (U5 was promoted in the 2026-04-26 cycle); the present paper documents and formalizes the result for the publication-series record.

1 Introduction

1.1 What U5 is

U5 is the structural commitment that the participation-measure Hilbert space $\mathcal{H}$ — established by the U2-Discrete and U2-Continuum theorems — carries a strongly continuous one-parameter unitary representation of spatial translations whose unique self-adjoint generator is the momentum operator $\hat{p}$. Equivalently, U5 supplies the kinematic translation-conjugate of position and identifies it as the operator whose eigenfunctions are plane waves and whose position-representation form is $\hat{p} = -i\hbar\nabla$.

The earlier framing of U5 in the QM-emergence Phase-1 inventory was as the *adjacency-band Fourier-conjugate partition*, $b^{\mathrm{adj}} = b_x + b_p$. The two framings are equivalent: the partition exists, is two-component, is Fourier-conjugate, and is orthogonal exactly when the translation group on $\mathcal{H}$ admits a unique self-adjoint generator and the standard L^2 Fourier transform supplies the basis-change between its position and momentum eigenbases. The present paper adopts the translation-symmetry framing because it makes the U5 result structurally transparent: U5 is, at root, *Stone's theorem applied to translation symmetry on the U2 Hilbert space.*

1.2 Why U5 matters

Three of the four foundational postulates of non-relativistic single-particle quantum mechanics — the Born rule, the Bell–Tsirelson bound, and the Heisenberg uncertainty inequality — depend on U5 either directly or via the inner-product structure that U5's translation operators preserve. The fourth foundational postulate — Schrödinger evolution — depends on U5 indirectly: the Hamiltonian $\hat{H} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$ derived in U4 contains the momentum operator that U5 supplies.

U5 is therefore the *kinematic backbone* of the structural-foundations program. U2 builds the Hilbert space; Theorem 10 (Born) installs the probabilistic interpretation; U5 supplies the translation generator and the position-momentum conjugacy; U1 derives the participation-measure that lives in the space; U4 derives the specific Hamiltonian form that acts on it. U3, the sole remaining upstream CANDIDATE, supplies the time-evolution generator's existence + linear-first-order structure; U5 is its spatial-translation analog and methodologically prefigures it.

1.3 Relationship to U1, U2, U4, and U3

- **U1.** U5 takes the participation-measure construction $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$ as input. The polarity-phase factor $e^{i\pi_K}$ supplies the $U(1)$-valued phase whose gradient identifies the momentum-conjugate direction.
- **U2.** U5 takes the U2 Hilbert space and its sesquilinear inner product as input. Translation operators are unitary on the U2 Hilbert space; Stone's theorem applies.
- **U4.** U5 supplies the momentum operator $\hat{p}$ that appears in U4's Hamiltonian form. U4's Galilean Lie algebra integration acts on $\hat{p}$ and would not even be statable without it.
- **U3.** U5 is *strictly independent* of U3. The U5 derivation invokes no time-evolution operator, no Hamiltonian, no Schrödinger equation. This is the central methodological feature of the present derivation and the principal sharpening over the earlier 2026-04-24 tightening-program memo, which had derived the Fourier conjugacy via U3.

1.4 Summary of the U5 verdict

The U5 verdict is **forced**. The three sub-features F1, F2, F3 close uniquely under the available structural inputs. All four falsifiers (Fal-1 through Fal-4) dispatch. No new structural CANDIDATE is introduced. The derivation is U3-independent. The U5 verdict is unconditional in both the discrete and continuum regimes; the U2-Continuum conformal gauge acts trivially on the translation representation and preserves the partition structure pointwise.

1.5 Organization

Section 2 states the U5 question precisely and inventories its sub-features and falsifiers. Section 3 inventories the structural inputs and the inputs explicitly forbidden under the U3-independence discipline. Section 4 decomposes U5 into F1, F2, F3 with status assignments. Section 5 introduces the four falsifiers. Sections 6–8 derive F1, F2, F3 in turn. Section 9 consolidates the falsifier dispatches. Section 10 documents the conditionality ledger. Section 11 states the final U5 theorem. Section 12 documents the downstream cascade. Section 13 discusses the result's significance and identifies the U3 arc as the natural successor.

2 Statement of the U5 Question

2.1 The target structure

The target structural content of U5 is the existence and uniqueness of the self-adjoint generator of spatial translations on the U2 Hilbert space, together with its identification as the momentum operator with the standard plane-wave eigenfunctions.

Formally: let $\mathcal{H}$ denote the U2 Hilbert space of participation measures (discrete or continuum regime, gauge-class equivalent under the U2-Continuum conformal redundancy). Let T_a for $a \in \mathbb{R}^d$ denote the spatial translation operator $T_a P(x) = P(x - a)$ acting componentwise on the participation-measure components. The U5 question is whether $\{T_a : a \in \mathbb{R}^d\}$ is a strongly continuous one-parameter unitary group whose unique self-adjoint generator is the standard momentum operator $\hat{p} = -i\hbar\nabla$ in the position representation, with plane waves $\langle x|k\rangle = (2\pi\hbar)^{-d/2} e^{ikx/\hbar}$ as eigenfunctions.

2.2 The three sub-features

- **F1 (Existence of translation representation).** $\{T_a : a \in \mathbb{R}^d\}$ is a well-defined, strongly continuous, one-parameter (per spatial axis) unitary group on $\mathcal{H}$.
- **F2 (Uniqueness of the self-adjoint generator).** There exists a unique self-adjoint operator $\hat{p}$ on $\mathcal{H}$ such that $T_a = \exp(i\hat{p} \cdot a/\hbar)$.
- **F3 (Identification with the momentum operator).** In the position representation of the U2 Hilbert space, $\hat{p} = -i\hbar\nabla$, with eigenfunctions $\langle x|k\rangle = (2\pi\hbar)^{-d/2} e^{ikx/\hbar}$, and the unique unitary intertwiner of position and momentum representations is the standard L^2 Fourier transform.

2.3 The four falsifiers

- **Fal-1 (Non-unitary translations).** T_a acts on $\mathcal{H}$ but does not preserve the U2 inner product.
- **Fal-2 (Multiple inequivalent generators).** The translation group admits more than one self-adjoint generator (Stone's theorem fails on $\mathcal{H}$).
- **Fal-3 (Non-self-adjoint generator).** A generator exists but is not self-adjoint, precluding interpretation as a physical observable.
- **Fal-4 (Non-linear translation action).** The translation action on participation-measure components is non-linear, excluding representation-theoretic treatment.

2.4 Methodological framing: U3 forbidden

The U5 arc adopts the discipline that **U3 may not appear in the derivation in any role**: not as a derivation input, not as an existence assumption, not as scope conditioning. This is *not* Framing 1 (which appeared in U4 to handle a sibling-CANDIDATE sibling-conditionality structure). U5 is structurally upstream of U3: the spatial translation generator must be installed before the time-translation generator can be identified as its temporal analog (which is U3's content). To derive U5 conditional on U3 would invert the dependency and re-introduce the methodological flaw of the 2026-04-24 tightening-program memo.

The forbidden-input discipline is enforced by an explicit circularity audit (§3.5).

3 Structural Inputs

3.1 Primitives

- **Primitive 03 (Participation):** graph homogeneity. The participation relation does not privilege any vertex of the participation graph. This supplies translation symmetry as a kinematic feature of the graph.
- **Primitive 06 (ED Gradient):** $\nabla\rho$ supplies a continuous spatial axis. Translations along this axis are well-defined and admissible. No preferred origin.
- **Primitive 09 (Tension Polarity):** $U(1)$-valued polarity phase $\pi(K, x)$. The gradient $\nabla\pi$ identifies the phase-propagation rate k, which is the wavenumber Fourier-conjugate to position.
- **Primitive 04 (Bandwidth):** supplies the four-band decomposition; the adjacency band b^{adj} is the band on which the F1–F3 partition acts.

3.2 Inherited forced upstream items

The U5 arc has access to the following theorem-grade upstream items established in the prior arc cycle:

- **U2-Discrete (Theorem 11).** Sesquilinear inner product on the discrete participation-measure Hilbert space.
- **U2-Continuum (Theorem 12).** Continuum lift of the U2 inner product, with explicit conformal gauge structure.
- **Theorem 10 (Born rule, Gleason–Busch).** Bandwidth-density identifications $b_x \propto |\Psi(x)|^2$ and $b_p \propto |\widetilde{\psi}(p)|^2$ inherit from Theorem 10 applied per band.

3.3 Participation-graph kinematics

The participation graph $G = (V, E)$ supplies vertices (Primitive 01), edges (Primitive 03), and bandwidth on the edges (Primitive 04). The translation group $\{T_a : a \in \mathbb{R}^d\}$ acts on participation-measure components by $T_a P_K(x) = P_K(x - a)$. This action is well-defined on $\mathcal{H}$ at the kinematic level — no dynamical evolution required.

3.4 Mathematical infrastructure

- Stone's theorem on one-parameter unitary groups (Stone 1932) [8].
- Standard L^2 Fourier transform on $L^2(\mathbb{R}^d)$, with Parseval's identity and the inversion formula [10].
- Standard functional calculus on self-adjoint operators.

3.5 Forbidden inputs (circularity audit)

The following inputs are explicitly forbidden as derivation inputs to U5:

- **U3** (the participation-measure evolution equation $i\hbar\partial_t P_K = \hat{H} P_K + \cdots$). Forbidden in any role.
- **The Schrödinger equation in textbook form.** Forbidden as a derivation input.

- **Any time-evolution operator or Hamiltonian.** Specifically, $\hat{H}$ does not appear in the U5 derivation.
- **U4** (the specific Hamiltonian form). Forbidden, since U4 is structurally downstream of U5.

Each subsequent derivation (§§6–8) operates strictly within the primitive-input scope of the present arc and references the forbidden-input list explicitly to confirm acyclicity. This audit follows the canonical templates of U1 Memo 01 §4 and U4 Memo 01 §2.

3.6 Scope conditions

- **Non-relativistic single-particle scope.** Inherited from Phase-1 Step 2's framing. The U5 derivation does not assume relativistic kinematics; the translation group is the abelian $\mathbb{R}^d$ group, not a Galilean or Lorentzian boost group (those enter U4).
- **Spatial-translation scope.** The U5 arc treats spatial translations only. Time translations (which would invoke Primitive 13 in addition to Primitives 03, 06) are U3's content.

4 Sub-Feature Decomposition

Sub-feature	Status	Structural inputs	Falsifiers
F1 (translation representation)	FORCED	Primitives 03 + 06 + U2 inner product	Fal-1, Fal-4
F2 (uniqueness of generator)	FORCED	F1 + Stone's theorem	Fal-2, Fal-3
F3 (momentum identification)	FORCED	F1 + F2 + U1 polarity-phase + standard L^2 Fourier theory	(consequence of F1–F2; structural specificity via dismissal of fractional-Fourier / wavelet / Mellin)

All three sub-features are forced-via-derivation. None is automatic; none is load-bearing in the sense of requiring a fresh primitive-level commitment. The closure of F3 requires explicit dismissal of three classes of alternative conjugacies (§8.5).

5 Falsifier Inventory

Each falsifier targets a specific physical alternative and a specific failure mode of the U5 commitment.

- **Fal-1 (Non-unitary translations).** Spatial translations exist as set-theoretic operators on $\mathcal{H}$ but fail to preserve the U2 inner product. Physical interpretation: "translating" a state would not preserve probability amplitudes, contradicting the homogeneity of physical space. Resolved in §6 by direct verification that T_a preserves the U2 inner product.
- **Fal-2 (Multiple inequivalent generators).** The translation group admits two or more inequivalent self-adjoint generators on $\mathcal{H}$, leaving the momentum operator structurally ambiguous. Physical interpretation: there would be no unique momentum observable, undermining its operational meaning. Resolved in §7 by Stone's theorem.

- **Fal-3 (Non-self-adjoint generator).** A generator exists but is not self-adjoint. Physical interpretation: momentum would not be a physical observable; eigenvalues need not be real. Resolved in §7: Stone's theorem produces a self-adjoint generator from a strongly continuous unitary group.
- **Fal-4 (Non-linear translation action).** Translations do not act linearly on participation-measure components, precluding representation-theoretic treatment. Physical interpretation: superposition would not be preserved under translation. Resolved in §6 by linearity of the translation action $T_a P_K(x) = P_K(x-a)$.

6 Derivation of F1 (Translation Representation)

6.1 The claim

The spatial translation group $\{T_a : a \in \mathbb{R}^d\}$ is a well-defined, strongly continuous, abelian d-parameter unitary group on the U2 Hilbert space $\mathcal{H}$.

6.2 Translation symmetry as a kinematic feature

Define the translation operator T_a acting componentwise on participation-measure components:

$$(T_a P)_K(x) := P_K(x-a), \qquad a \in \mathbb{R}^d. \tag{1}$$

Three primitive-level observations:

(i) Primitive 03's graph-homogeneity guarantees that no vertex is privileged; the participation relation is invariant under graph automorphisms induced by spatial translations.

(ii) Primitive 06's gradient axis $\nabla\rho$ is continuous and unbounded; translations along this axis are well-defined for all $a \in \mathbb{R}^d$.

(iii) No primitive-level structure singles out a privileged origin or a privileged translation magnitude.

These three observations together establish translation symmetry as a *kinematic* property of the participation graph — a symmetry of the static structure, not of any dynamical evolution. **No reference to U3, to time evolution, or to a Hamiltonian appears in this argument.**

6.3 Linearity

Translation acts linearly:

$$T_a(\alpha P + \beta Q)_K(x) = \alpha P_K(x-a) + \beta Q_K(x-a) = \alpha (T_a P)_K(x) + \beta (T_a Q)_K(x)$$

for all $\alpha, \beta \in \mathbb{C}$ and all $P, Q \in \mathcal{H}$. **Falsifier Fal-4 dispatched.**

6.4 Unitarity

For $P, Q \in \mathcal{H}$ and $a \in \mathbb{R}^d$:

$$\langle T_a P \mid T_a Q \rangle_{\mathcal{H}} = \sum_K \int_{\mathbb{R}^d} P_K^*(x-a)\, Q_K(x-a)\, d\mu(x).$$

Substituting $x' = x - a$, with $d\mu(x') = d\mu(x)$ since the U2 measure is translation-invariant (Primitive 06 supplies a translation-invariant Lebesgue measure on the spatial axis; the U2-Continuum gauge acts as a conformal rescaling that does not depend on x in the flat baseline):

$$\langle T_a P \mid T_a Q \rangle_{\mathcal{H}} = \sum_K \int_{\mathbb{R}^d} P_K^*(x')\, Q_K(x')\, d\mu(x') = \langle P \mid Q \rangle_{\mathcal{H}}.$$

The translation operators preserve the U2 inner product. **Falsifier Fal-1 dispatched.**

6.5 Group structure

Direct calculation: $T_a T_b = T_{a+b}$, $T_0 = I$, $T_{-a} = T_a^{-1}$. The translation group is abelian and isomorphic to $(\mathbb{R}^d, +)$.

6.6 Strong continuity

For any $P \in \mathcal{H}$ with components P_K in $L^2(d\mu)$, the map $a \mapsto T_a P$ is strongly continuous because L^2-translation is strongly continuous. (This is a standard property of L^2, inherited by the direct sum / direct integral structure of the U2 Hilbert space across the channel index K.)

Theorem 6.1 (F1 — Translation Representation)**.** *Let $\mathcal{H}$ denote the U2 Hilbert space of participation measures established by the U2-Discrete and U2-Continuum theorems. Then the spatial translation operators $T_a P_K(x) := P_K(x-a)$ for $a \in \mathbb{R}^d$ form a strongly continuous abelian d-parameter unitary group on $\mathcal{H}$, derived from Primitives 03 and 06 plus the U2 inner-product structure, with no input from U3 or any time-evolution operator.*

7 Derivation of F2 (Uniqueness of the Generator)

7.1 Stone's theorem

Theorem 7.1 (Stone 1932 [8])**.** *Every strongly continuous one-parameter unitary group $\{U(t) : t \in \mathbb{R}\}$ on a Hilbert space has a unique self-adjoint generator A such that $U(t) = \exp(itA)$.*

7.2 Application to F1's translation group

By Theorem F1, $\{T_a : a \in \mathbb{R}^d\}$ is a strongly continuous unitary group on $\mathcal{H}$. Apply Stone's theorem to each spatial direction $\hat{e}_i$ with $a = a_i \hat{e}_i$: there exists a unique self-adjoint generator $\hat{p}_i$ on $\mathcal{H}$ such that

$$T_{a_i \hat{e}_i} = \exp\left(\frac{i \hat{p}_i a_i}{\hbar}\right).$$

The d generators $\hat{p}_i$ commute pairwise (by the abelian structure of the translation group, $T_a T_b = T_{a+b} = T_b T_a$), so the joint exponentiation is well-defined:

$$T_a = \exp\left(\frac{i \hat{p} \cdot a}{\hbar}\right), \qquad \hat{p} := (\hat{p}_1, \ldots, \hat{p}_d). \tag{2}$$

The generator $\hat{p}$ is unique. **Falsifier Fal-2 dispatched.**

7.3 Self-adjointness

Stone's theorem produces a self-adjoint generator from a strongly continuous unitary group. The self-adjointness is automatic from the theorem and does not require separate verification. **Falsifier Fal-3 dispatched.**

7.4 Why this does not require U3

Stone's theorem is a statement about strongly continuous one-parameter unitary groups on Hilbert spaces. The translation group $\{T_a\}$ is one such group on $\mathcal{H}$. The hypotheses of Stone's theorem are entirely kinematic: the existence of $\mathcal{H}$ (supplied by U2), the unitarity of T_a (Theorem F1), and the strong continuity of T_a (Theorem F1). At no point does the theorem require the existence of a *time*-evolution operator, a Hamiltonian, or any dynamical input. The translation generator $\hat{p}$ is identified as a *spatial* self-adjoint operator on $\mathcal{H}$ at fixed time, with no reference to the participation-measure evolution equation.

Theorem 7.2 (F2 — Uniqueness of the Translation Generator)**.** *Under the hypotheses of Theorem F1, there exists a unique self-adjoint operator $\hat{p}$ on $\mathcal{H}$ such that $T_a = \exp(i\hat{p} \cdot a/\hbar)$ for all $a \in \mathbb{R}^d$. The identification is via Stone's theorem applied to the strongly continuous unitary group $\{T_a\}$, and is U3-independent.*

8 Derivation of F3 (Momentum Operator Identification)

8.1 The position-representation form

In the position representation of $\mathcal{H}$, the translation operator acts on a wavefunction $\psi(x)$ by $T_a\psi(x) = \psi(x-a)$. Taylor-expansion in a for analytic ψ:

$$T_a\psi(x) = \psi(x-a) = \sum_{n=0}^{\infty} \frac{(-a\cdot\nabla)^n}{n!}\psi(x) = \exp(-a\cdot\nabla)\psi(x). \tag{3}$$

Comparing with (2):

$$\exp\left(\frac{i\hat{p}\cdot a}{\hbar}\right) = \exp(-a\cdot\nabla)$$

which is satisfied by

$$\boxed{\hat{p} = -i\hbar\nabla.} \tag{4}$$

8.2 Plane-wave eigenfunctions

The eigenvalue equation $\hat{p}\,|k\rangle = k|k\rangle$ in the position representation reads $-i\hbar\nabla\langle x|k\rangle = k\langle x|k\rangle$, with solution

$$\langle x|k\rangle = (2\pi\hbar)^{-d/2}\, e^{ik\cdot x/\hbar}. \tag{5}$$

The normalization $(2\pi\hbar)^{-d/2}$ is fixed by the orthonormality $\langle k|k'\rangle = \delta(k-k')$ via the standard Fourier convention.

8.3 The thin-limit / continuum identification

In the discrete regime, the channel index K runs over a discrete set; in the continuum / thin limit, K is replaced by a continuous wavenumber $k \in \mathbb{R}^d$. The identification is:

$$K \xleftrightarrow{\text{thin limit}} k. \tag{6}$$

The U1 polarity-phase factor $e^{i\pi_K(x)}$ in the participation-measure construction $P_K = \sqrt{b_K}\, e^{i\pi_K}$ is what makes this identification natural: in the continuum limit, $\pi_K(x) \to k\cdot x/\hbar$, so the polarity phase is precisely the plane-wave phase. The U1-supplied phase structure is the structural bridge between the channel index and the momentum eigenvalue.

8.4 The role of the U2 inner product

The U2 inner product $\langle P|Q\rangle = \sum_K \int P_K^* Q_K \, d\mu$ supplies the Hilbert-space structure on which:

(i) The Fourier transform $\widetilde{\psi}(k) = (2\pi\hbar)^{-d/2} \int e^{-ikx/\hbar}\psi(x)\, dx$ is an isometry: $\langle\psi|\phi\rangle_x = \langle\widetilde{\psi}|\widetilde{\phi}\rangle_k$ (Parseval).

(ii) The position eigenstates $\{|x\rangle\}$ and momentum eigenstates $\{|k\rangle\}$ are both complete orthonormal bases.

(iii) The standard Fourier transform is the unique unitary intertwiner between the two bases.

8.5 Dismissal of alternative conjugacies

Three classes of alternative integral transforms might in principle have served as the conjugacy between position and momentum representations. Each is dismissed:

- **Fractional-Fourier transforms.** Diagonalize rotated phase-space operators (specifically, the operator $\cos\theta \cdot \hat{x} + \sin\theta \cdot \hat{p}$ for fractional angle θ). Not the translation generator $\hat{p}$. Fail to provide the *translation*-conjugate basis. Inconsistent with the Stone's-theorem identification.
- **Wavelet transforms.** Multi-scale resolutions; not Stone's-theorem intertwiners of any one-parameter unitary group. Provide a different mathematical structure (overcomplete frames, time-frequency tilings) that is not the translation-conjugate basis.
- **Mellin transforms.** Intertwine the dilation group $\{x \to e^t x\}$, not the translation group. Dilations are not a primitive-level kinematic symmetry of the participation graph (the participation graph has no preferred scale; dilation symmetry is not supplied by any primitive). Mellin transforms therefore do not arise structurally.

The standard L^2 Fourier transform is the **unique** unitary intertwiner of position and translation-conjugate momentum representations. It is forced by (a) the translation symmetry of the participation graph, (b) the Stone's-theorem identification of the unique self-adjoint generator, and (c) the standard L^2 Fourier theory.

Theorem 8.1 (F3 — Momentum Operator Identification)**.** *Let $\hat{p}$ be the unique self-adjoint translation generator on the U2 Hilbert space $\mathcal{H}$ supplied by Theorem F2. Then in the position representation of $\mathcal{H}$, $\hat{p} = -i\hbar\nabla$. The eigenfunctions of $\hat{p}$ are plane waves $\langle x|k\rangle = (2\pi\hbar)^{-d/2}\, e^{ikx/\hbar}$. The unique unitary intertwiner of position and momentum representations is the standard L^2 Fourier transform; fractional-Fourier, wavelet, and Mellin alternatives are dismissed by structural inconsistency with the translation-driven conjugacy.*

9 Falsifier Resolution Table

Falsifier	Target	Dispatch argument	Section
Fal-1	F1	Direct calculation: $\langle T_a P \mid T_a Q\rangle = \langle P \mid Q\rangle$ via translation-invariance of the U2 measure	§6.4
Fal-2	F2	Stone's theorem on the strongly continuous unitary group $\{T_a\}$ produces a unique self-adjoint generator	§7.2
Fal-3	F2	Stone's theorem produces a self-adjoint generator from a strongly continuous unitary group	§7.3
Fal-4	F1	Direct linearity of T_a on participation-measure components	§6.3

All four falsifiers dispatched within the stated scope conditions and forbidden-input discipline.

10 Conditionality Ledger

The U5 verdict is **forced** with the following inherited conditionalities and zero new ones:

1. **Non-relativistic single-particle scope.** Inherited from Phase-1 Step 2; conditions the use of the abelian $(\mathbb{R}^d, +)$ translation group rather than a Galilean or Lorentzian boost group.

2. **U2 structure (inner product, both regimes).** Inherited from the U2-Discrete and U2-Continuum theorems; supplies the Hilbert space on which Stone's theorem applies.

3. **U1 participation-measure construction.** Inherited from the U1 theorem (FORCED); supplies the polarity-phase structure that natural-identifies the channel index with the momentum eigenvalue in the continuum limit (§8.3).

4. **Theorem 10 (Born) for downstream use only.** Used in the bandwidth-density identifications $b_x \propto |\Psi|^2$, $b_p \propto |\widetilde{\psi}|^2$ in the equivalence to the adjacency-band-partition framing; not used in the F1–F3 derivations themselves.

No dependence on U3. The U5 derivation invokes no time-evolution operator, no Hamiltonian, no Schrödinger equation. The translation symmetry is kinematic (Primitives 03 + 06), the unitary representation is established at fixed time on the U2 Hilbert space (Theorem F1), Stone's theorem identifies the generator without dynamical input (Theorem F2), and the position-representation form (Theorem F3) follows from comparison of operator exponentials. **U5 is structurally upstream of U3 and remains so under the present derivation.**

No new structural candidates introduced. All four conditionalities are inherited from prior structural commitments.

11 Final U5 Theorem

Theorem 11.1 (U5; Translation Symmetry and the Momentum Operator)**.** *Let the ED primitive stack supply: graph homogeneity (Primitive 03), bandwidth (Primitive 04), spatial gradient (Primitive 06), and tension polarity (Primitive 09). Let the participation-measure carrier (U1, FORCED) and the U2 inner product on the participation-measure Hilbert space (U2-Discrete + U2-Continuum, both FORCED) supply the Hilbert space $\mathcal{H}$. Then:*

1. ***Translation representation.*** *The spatial translation operators $T_a P_K(x) := P_K(x-a)$ for $a \in \mathbb{R}^d$ form a strongly continuous abelian d-parameter unitary group on $\mathcal{H}$.*

2. ***Unique self-adjoint generator.*** *By Stone's theorem, there exists a unique self-adjoint operator $\hat{p}$ on $\mathcal{H}$ such that*

$$T_a = \exp\left(\frac{i\hat{p}\cdot a}{\hbar}\right), \qquad a \in \mathbb{R}^d.$$

3. ***Momentum operator identification.*** *In the position representation of $\mathcal{H}$, $\hat{p} = -i\hbar\nabla$. The eigenfunctions of $\hat{p}$ are plane waves $\langle x|k\rangle = (2\pi\hbar)^{-d/2}e^{ikx/\hbar}$, with the standard L^2 Fourier transform as the unique unitary intertwiner of position and momentum representations.*

The derivation invokes no U3 input, no Hamiltonian, no time-evolution operator, and no Schrödinger equation. All four falsifiers (non-unitary translations, multiple inequivalent generators, non-self-adjoint generator, non-linear translation action) are dispatched.

Status: FORCED, unconditional in both discrete and continuum regimes; gauge-compatible with the U2-Continuum conformal redundancy (the gauge acts uniformly on both partition components and trivially on the translation representation); U3-independent.

12 Downstream Consequences

12.1 How U5 feeds into U4 (momentum identification → kinetic term)

U4's Hamiltonian form $\hat{H} = \hbar^2|\hat{p}|^2/(2m)+V(\hat{x})$ requires the prior identification of the momentum operator $\hat{p}$. U5 supplies $\hat{p}$ as a self-adjoint operator on the U2 Hilbert space with plane-wave eigenfunctions. U4's Galilean Lie algebra integration acts on this $\hat{p}$:

$$[\hat{H}, \hat{K}_i] = -i\hbar\hat{p}_i$$

with $\hat{K}_i = m\hat{x}_i - t\hat{p}_i$. The integration of this commutator condition (yielding $\partial T/\partial\hat{p}_i = \hat{p}_i/m$ and thus $T = |\hat{p}|^2/(2m)$) presupposes that $\hat{p}_i$ is a well-defined self-adjoint operator. **U5 supplies that operator.** Without U5, the U4 Galilean integration is not statable.

12.2 How U5 feeds into U1 (complex structure compatibility)

The U1 participation-measure construction $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$ supplies the polarity-phase factor $e^{i\pi_K}$ whose gradient identifies the wavenumber $k = \nabla\pi$. U5 closes the loop: the wavenumber k is the eigenvalue of the translation generator $\hat{p}$, and the complex-valued structure of the participation measure is precisely what makes the plane-wave eigenfunctions $e^{ikx/\hbar}$ available as a basis. The U1 complex structure and the U5 momentum operator are mutually compatible — neither was designed assuming the other, but both close consistently within the same Hilbert-space framework.

12.3 Updated structural-foundations ledger

Following the 2026-04-26 cycle, the inventory of structural-foundations theorems stands at fifteen forced items, with U5 occupying position #13:

#	Theorem	Status
10	Born rule (Gleason–Busch)	FORCED
11	U2-Discrete (sesquilinear inner product)	FORCED
12	U2-Continuum (with conformal gauge)	FORCED
13	**U5 (translation generator + momentum operator; this paper)**	**forced**
14	U1 (participation-measure construction)	FORCED
15	U4 (specific Hamiltonian form; form-FORCED–value-INHERITED)	FORCED cond. on U3 (Framin

12.4 Forced theorem count and active CANDIDATE inventory

The U5 paper formalizes a result already promoted in the 2026-04-26 cycle. The active upstream-CANDIDATE inventory therefore reads, post-cycle:

$$\{U3\} + \text{continuum gauge convention.}$$

One upstream CANDIDATE remains. Three of the four foundational QM postulates (Born rule, Bell–Tsirelson bound, Heisenberg uncertainty) are FORCED-unconditional. Schrödinger evolution is FORCED conditional on U3 alone.

13 Discussion and Outlook

13.1 Why U5 is the kinematic backbone of the structural-foundations program

The structural-foundations cycle decomposes into three layers:

- **Layer 1 (carrier):** U1 — the participation-measure construction.
- **Layer 2 (Hilbert-space arena):** U2-Discrete + U2-Continuum + Theorem 10 + **U5**.
- **Layer 3 (dynamical):** U3 + U4.

U5 occupies the Hilbert-space layer alongside U2 and Theorem 10. Within that layer, U5 is what installs the *kinematic* generators: spatial translations have a unique self-adjoint generator (the momentum operator), with plane-wave eigenfunctions and a well-defined Fourier-conjugate position basis. Without U5, the Hilbert space exists but carries no identified momentum observable; with U5, the kinematic skeleton of standard quantum mechanics is in place at fixed time.

The "kinematic backbone" framing is precise: U5 supplies everything needed to do single-time quantum kinematics — observables, eigenbases, conjugate-pair structure, Fourier conjugacy — without invoking time evolution. Time evolution is U3's content; the Hamiltonian's specific form is U4's content. U5 is the structural boundary between kinematics and dynamics.

13.2 How U5 + U2 + U4 jointly force the Schrödinger form

The dynamical content of non-relativistic single-particle quantum mechanics is the Schrödinger equation $i\hbar\partial_t\psi = \hat{H}\psi$ with $\hat{H} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$. Three structural commitments suffice to force it:

- **U2** supplies the Hilbert space.
- **U5** supplies the momentum operator $\hat{p}$ on that Hilbert space.
- **U4** supplies the Hamiltonian form $\hat{H} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$ via Galilean Lie algebra integration.

The remaining commitment — U3, the existence and linear-first-order structure of the time-evolution generator — is what binds these three pieces into a dynamical equation. With U3 closed, the Schrödinger equation is structurally complete: U2 supplies the arena, U5 supplies the kinematic generator, U4 supplies the dynamical generator's form, U3 supplies the existence and linearity of dynamical evolution.

The U5 derivation is what makes the U4 closure acyclic: U4's Galilean integration acts on the U5 momentum operator, and U5 was derived without invoking U4 or U3. Promoting U3 in the next arc would close the entire QM-emergence Phase-1 program.

13.3 Sharpening over earlier work

The earlier 2026-04-24 tightening-program memo `arcs/arc-foundations/u5_adjacency_partition_derivati` derived the Fourier conjugacy via U3 — invoking the participation-measure evolution equation as a derivation input. The present derivation removes that dependency: translation symmetry is established as a kinematic property of the participation graph (Primitives 03 + 06), Stone's theorem applies on the U2 Hilbert space without time-evolution input, and the Fourier conjugacy follows from the position-representation form of the translation generator together with the standard L^2 Fourier theory.

The methodological consequence is decoupling: U5's verdict is independent of whatever conditionality U3 carries. Promoting U3 in a future arc does not retroactively change the U5 result, and U5's promotion does not feed back into U3's derivation chain.

13.4 Next steps

1. **Open the U3 arc.** The natural and final foundational target. U3 inherits all six FORCED upstream items (U1, U2-Discrete, U2-Continuum, U5, Theorem 10, U4) and applies Stone's theorem to time-translation symmetry (Primitive 13) — paralleling F1 of the present paper applied to spatial translations. Headline result if it closes: all four foundational QM postulates structurally derived from ED primitives.

2. **Synthesis paper revision.** With U5 now formalized as a publication paper, the QM-emergence Phase-1 synthesis paper §§3.3/3.4/3.5/4 can be revised to reflect the four-paper publication series (Born_Gleason, U2_Inner_Product, U1_Participation_Measure, U4_Hamiltonian_Form) plus this paper. Recommend deferring the synthesis revision until U3 closes (single editorial pass).

3. **Public-facing U5 explainer.** A natural fifth in the public-explainer ladder, sitting between the U2 explainer (Hilbert space) and the U4 explainer (Hamiltonian form). The arc closure memo §6 supplies a draft.

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Manuscript prepared April 2026. Source materials: `arcs/U5/` four-memo arc and closure memo, formalized into a single publishable document. The U5 arc is the third of the structural-foundations

cycle (chronologically the fourth, drafted as the fifth publication paper after Born_Gleason, U2_Inner_Product, U1_Participation_Measure, and U4_Hamiltonian_Form).
Draft complete. Pausing for revision.